



**The CENTRE for EDUCATION
in MATHEMATICS and COMPUTING**

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**2023 Gauss Grade 8
Solutions**

1. Which of the following numbers is equivalent to the fraction $\frac{1}{4}$?
(A) 0.5 (B) 0.25 (C) 4.0 (D) 0.14 (E) 2.5

Source: 2023 Gauss Grade 8 #1

Primary Topics: Number Sense

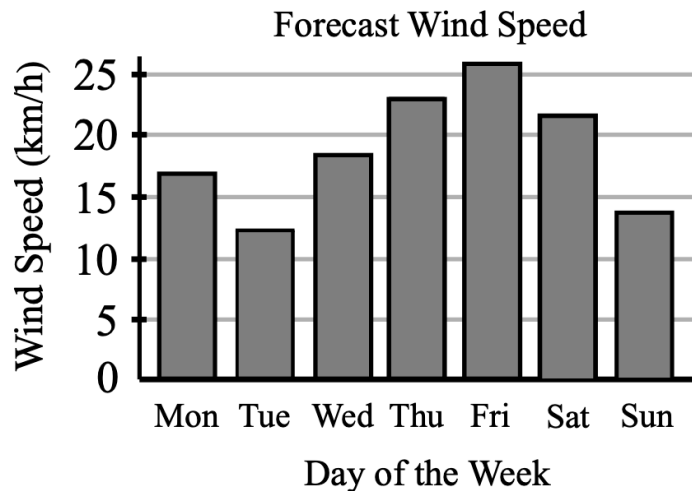
Secondary Topics: Fractions/Ratios | Counting

Answer: B

Solution:

The fraction $\frac{1}{4}$ is equivalent to $1 \div 4 = 0.25$.

2. The graph shows the forecast wind speed (in km/h) during a 7-day period.



Jack can sail alone only when the forecast wind speed is less than 20 km/h. During this 7-day period, on how many days will Jack be able to sail alone?

- (A) 4 (B) 6 (C) 1 (D) 2 (E) 5

Source: 2023 Gauss Grade 8 #2

Primary Topics: Data Analysis

Secondary Topics: Graphs | Counting

Answer: A

Solution:

Reading from the graph, the forecast wind speed is less than 20 km/h on Monday, Tuesday, Wednesday, and Sunday. Thus, Jack will be able to sail alone on 4 days during this 7-day period.

3. Which of the following numbers is *not* a multiple of 15?

- (A) 150 (B) 25 (C) 30 (D) 45 (E) 60

Source: 2023 Gauss Grade 8 #3

Primary Topics: Number Sense

Secondary Topics: Divisibility | Factoring | Counting

Answer: B

Solution:

We note that $15 \times 10 = 150$, $15 \times 2 = 30$, $15 \times 3 = 45$, and $15 \times 4 = 60$. Since there is no integer n for which $15 \times n = 25$, then 25 is not a multiple of 15.

4. If the integers $-7, 10, 9, 0, -9$ are ordered from least to greatest, what is the third integer in the list?

- (A) -7 (B) 10 (C) 9 (D) 0 (E) -9

Source: 2023 Gauss Grade 8 #4

Primary Topics: Number Sense

Secondary Topics: Counting

Answer: D

Solution:

Ordering the given list of integers from least to greatest, we get $-9, -7, 0, 9, 10$. The third integer in the ordered list is 0.

5. If $2n = 14$, the value of $10n$ is

- (A) 14 (B) 140 (C) 70 (D) 28 (E) 56

Source: 2023 Gauss Grade 8 #5

Primary Topics: Number Sense | Algebra and Equations

Secondary Topics: Equations Solving | Exponents | Expressions

Answer: C

Solution:

Solution 1

If $2n = 14$, then $n = \frac{14}{2} = 7$. When $n = 7$, the value of $10n$ is $10 \times 7 = 70$.

Solution 2

Multiplying both sides of the given equation by 5, we get $5 \times 2n = 5 \times 14$, and so $10n = 70$.

6. Tallulah is playing a game in which she rolls a single standard die once. If the number rolled is 1, 2, 3, or 4, then she wins. If the number rolled is 5 or 6, then she loses. What is the probability that she loses?

- (A) $\frac{1}{3}$ (B) $\frac{5}{6}$ (C) $\frac{1}{2}$ (D) $\frac{1}{6}$ (E) $\frac{11}{12}$

Source: 2023 Gauss Grade 8 #6

Primary Topics: Other | Counting and Probability

Secondary Topics: Games | Counting | Probability | Fractions/Ratios

Answer: A

Solution:

There are 6 possible outcomes when Tallulah rolls a single standard die once. She loses if she rolls 2 of these 6 outcomes, and so the probability that she loses is $\frac{2}{6} = \frac{1}{3}$.

7. In the addition shown, P and Q are each equal to a digit.

$$\begin{array}{r} 1013 \\ + PQPQ \\ \hline 2023 \end{array}$$

The value of $P + Q$ is

- (A) 4 (B) 1 (C) 0 (D) 3 (E) 5

Source: 2023 Gauss Grade 8 #7

Primary Topics: Number Sense

Secondary Topics: Counting | Digits

Answer: B

Solution:

Solution 1

We may convert the given addition problem to a subtraction problem. That is, since $1013 + PQPQ = 2023$, then $PQPQ = 2023 - 1013$. The difference between 2023 and 1013 is $2023 - 1013 = 1010$, and so $P = 1$, $Q = 0$, and $P + Q = 1 + 0 = 1$.

Solution 2

The ones (units) digit of the sum 2023 is 3. Thus, the ones digit of the sum $3 + Q$ must equal 3. Since Q is a digit, the only possible value of Q is 0. The tens digit of the sum 2023 is 2. Since there is no "carry" from the ones column to the tens column, the ones digit of the sum $1 + P$ must equal 2. Since P is a digit, the only possible value of P is 1. We may confirm that when $P = 1$ and $Q = 0$, we get $1013 + 1010 = 2023$ as required. The value of $P + Q$ is $1 + 0 = 1$.

8. In a salad dressing, the ratio of oil to vinegar is 3 : 1. To make the dressing more acidic, the amount of vinegar is doubled. What is the new ratio of oil to vinegar?
- (A) 3 : 2 (B) 6 : 1 (C) 1 : 3 (D) 2 : 3 (E) 4 : 3

Source: 2023 Gauss Grade 8 #8

Primary Topics: Number Sense

Secondary Topics: Fractions/Ratios | Expressions

Answer: A

Solution:

Suppose the salad dressing initially contains 300 mL of oil. Since the ratio of oil to vinegar is 3 : 1, then the salad dressing initially contains one-third as much vinegar as oil, or $\frac{1}{3} \times 300 \text{ mL} = 100 \text{ mL}$ of vinegar (note that $300 : 100 = 3 : 1$).

If the volume of vinegar is doubled, the new volume of vinegar is $2 \times 100 \text{ mL} = 200 \text{ mL}$, and so the new ratio of oil to vinegar is $300 : 200 = 3 : 2$. Note: We chose to begin with 300 mL of oil, but performing the above calculations with any starting volume of oil will give the same 3 : 2 ratio.

9. A grocery receipt shows the cost of three items before tax is added.

Sponge	\$4.20
Shampoo	\$7.60
Soap	\$3.20

A 5% tax is added to the cost of the items. The total cost of the three items, including tax, is

- (A) \$15.16 (B) \$15.08 (C) \$15.22 (D) \$15.75 (E) \$15.38

Source: 2023 Gauss Grade 8 #9

Primary Topics: Other | Algebra and Equations

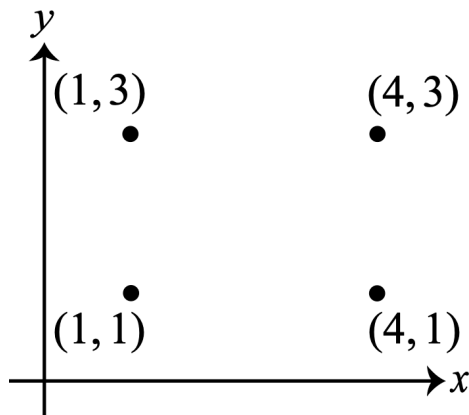
Secondary Topics: Percentages | Expressions | Rates

Answer: D

Solution:

Before including tax, the combined cost of the three items is $\$4.20 + \$7.60 + \$3.20 = \15.00 . The 5% tax on \$15.00 is $0.05 \times \$15.00 = \0.75 , and so the total cost of the three items, including tax, is $\$15.00 + \$0.75 = \$15.75$. Note that we could have calculated the 5% tax on each individual item, however doing so is less efficient than calculating tax on the \$15.00 total.

10. The vertices of a rectangle have coordinates $(1, 3)$, $(1, 1)$, $(4, 1)$, and $(4, 3)$, as shown.



If the rectangle is reflected in the y -axis, which of the following points is *not* a vertex of the reflected rectangle?

- (A) $(-1, 1)$ (B) $(-4, 1)$ (C) $(-3, 4)$ (D) $(-1, 3)$ (E) $(-4, 3)$

Source: 2023 Gauss Grade 8 #10

Primary Topics: Data Analysis

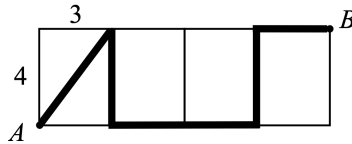
Secondary Topics: Graphs | Coordinate Geometry | Transformations

Answer: C

Solution:

When $(1, 3)$ is reflected in the y -axis, the reflected point is $(-1, 3)$. In general, when a point is reflected in the y -axis, its x -coordinate changes sign, and its y -coordinate does not change. Thus, the vertices of the reflected rectangle are $(-1, 3)$, $(-1, 1)$, $(-4, 1)$, and $(-4, 3)$. Of the given possibilities, $(-3, 4)$ is not a vertex of the reflected rectangle.

11. The diagram is made up of four congruent rectangles with dimensions 3 by 4.



The four rectangles are arranged side-by-side forming one larger rectangle. The left side of the large rectangle is labeled 4. The top (and bottom) is the total of four sides of length 3. A is the bottom left vertex of the large rectangle and B is the top right corner. A path is made up of line segments. It starts at A, moves diagonally up across the first rectangle, down the side shared by rectangles 1 and 2, across the bottoms of rectangles 2 and 3, up the shared side of rectangles 3 and 4, and the across the top of rectangle 4, ending at B.

What is the length of the path from A to B shown on the diagram?

- (A) 22 (B) 21 (C) 19 (D) 20 (E) 23

Source: 2023 Gauss Grade 8 #11

Primary Topics: Geometry and Measurement

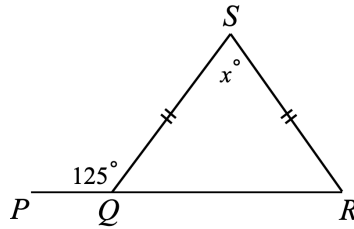
Secondary Topics: Quadrilaterals | Measurement | Pythagorean Theorem

Answer: A

Solution:

In the leftmost rectangle, the length of the path along the rectangle's diagonal, d , and the sides with lengths 3 and 4, form a right-angled triangle. Using the Pythagorean Theorem, we get $d^2 = 3^2 + 4^2$, and so $d = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$ (these are "3-4-5" right-angled triangles). The path from A to B includes one such diagonal, two vertical sides each of length 4, and three horizontal sides each of length 3, and thus has length $5 + (2 \times 4) + (3 \times 3) = 22$.

12. In the diagram, PQR is a line segment, $\angle PQS = 125^\circ$, $\angle QSR = x^\circ$, and $SQ = SR$.



What is the value of x ?

- (A) 60 (B) 70 (C) 80 (D) 110 (E) 125

Source: 2023 Gauss Grade 8 #12

Primary Topics: Geometry and Measurement

Secondary Topics: Angles | Expressions | Triangles | Measurement

Answer: B

Solution:

Since $\angle PQR$ is a straight angle, its measure is 180° , and so $\angle SQR = 180^\circ - 125^\circ = 55^\circ$. Since $SQ = SR$, then $\angle SRQ = \angle SQR = 55^\circ$. The sum of the angles inside $\triangle SQR$ is 180° , and so $x = 180 - 55 - 55 = 70$.

13. When attempting to arrange a pile of peaches into groups of threes, there are two peaches not in a group of three. Which of the following choices could be the number of peaches in the original pile?

- (A) 19 (B) 49 (C) 33 (D) 29 (E) 61

Source: 2023 Gauss Grade 8 #13

Primary Topics: Counting and Probability | Number Sense

Secondary Topics: Counting | Divisibility

Answer: D

Solution:

The number of peaches in the original pile is two more than a multiple of three. Of the choices given, 29 is the only number which is two more than a multiple of three ($29 = 3 \times 9 + 2$).

14. A list of 5 integers repeats to form the pattern:

$$4, -3, 2, -1, 0, 4, -3, 2, -1, 0, \dots$$

What is the sum of the first 23 integers?

- (A) 3 (B) 8 (C) 10 (D) 11 (E) 13

Source: 2023 Gauss Grade 8 #14

Primary Topics: Algebra and Equations

Secondary Topics: Patterning/Sequences/Series | Counting | Expressions

Answer: D

Solution:

The sum of each block of 5 repeating integers is $4 - 3 + 2 - 1 + 0 = 2$. In the first 23 integers, there are 4 such blocks of 5 integers, plus 3 additional integers (since $23 = 4 \times 5 + 3$). The sum of the 4 blocks of 5 repeating integers is $4 \times 2 = 8$, and the next three integers in the list following these 4 blocks are 4, -3, 2. Thus, the sum of the first 23 integers is $8 + 4 - 3 + 2 = 11$.

15. Bindu's bike tires have a radius of 30 cm. She rides her bike far enough that the tires rotate exactly five times. How far does Bindu's bike travel?

- (A) 60π cm (B) 30π cm (C) 900π cm (D) 300π cm (E) 150π cm

Source: 2023 Gauss Grade 8 #15

Primary Topics: Geometry and Measurement

Secondary Topics: Circles | Measurement | Expressions

Answer: D

Solution:

The circumference of each of Bindu's bike tires is $2 \times \pi \times 30 \text{ cm} = 60\pi \text{ cm}$. If the bike tires rotate exactly five times, then the distance travelled by Bindu's bike is $5 \times 60\pi \text{ cm} = 300\pi \text{ cm}$.

16. The numbers 41, 35, 19, 9, 26, 45, 13, 28 are arranged in pairs so that the sum of the numbers in each pair is the same. The number paired with 13 is
 (A) 45 (B) 28 (C) 35 (D) 26 (E) 41

Source: 2023 Gauss Grade 8 #16

Primary Topics: Number Sense | Algebra and Equations

Secondary Topics: Patterning/Sequences/Series | Counting | Operations | Expressions

Answer: E

Solution:

Solution 1

The sum of all 8 numbers in the list is $41 + 35 + 19 + 9 + 26 + 45 + 13 + 28 = 216$. When the 8 numbers are arranged in pairs, there are 4 pairs. The sum of the numbers in each pair is the same, and so this sum is $\frac{216}{4} = 54$. Therefore, the number paired with 13 is $54 - 13 = 41$. Note: We may confirm that $45 + 9 = 54$, $41 + 13 = 54$, $35 + 19 = 54$, and $28 + 26 = 54$.

Solution 2

Since the sum of the numbers in each pair is the same, then the largest number in the list must be paired with the smallest number, the second largest with the second smallest, and so on. (Can you reason why this must be true?) That is, the largest and smallest numbers in the list, 45 and 9, must be paired. The second largest and second smallest numbers, 41 and 13, must be paired, and so the number paired with 13 is 41. Note: We may confirm that $45 + 9 = 54$, $41 + 13 = 54$, $35 + 19 = 54$, and $28 + 26 = 54$.

17. For 30 consecutive days, the daily high temperature was recorded. On each of the first 25 days, the temperature recorded was 21°C . On each of the remaining 5 days, the temperature recorded was 15°C . For the 30 days, the mean (average) of the temperatures recorded was
 (A) 17°C (B) 19°C (C) 20°C (D) 16°C (E) 18°C

Source: 2023 Gauss Grade 8 #17

Primary Topics: Data Analysis

Secondary Topics: Averages | Counting | Expressions

Answer: C

Solution:

The mean (average) is determined by adding the 30 recorded temperatures, and dividing the sum by 30. The sum of the temperatures for the first 25 days was $(25 \times 21)^{\circ}\text{C} = 525^{\circ}\text{C}$. The sum of the temperatures for the last 5 days was $(5 \times 15)^{\circ}\text{C} = 75^{\circ}\text{C}$. Thus, the mean of the recorded temperatures was $\frac{525^{\circ}\text{C} + 75^{\circ}\text{C}}{30} = \frac{600^{\circ}\text{C}}{30} = 20^{\circ}\text{C}$.

18. The product of a pair of 2-digit positive integers is 630. How many such pairs are there?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Source: 2023 Gauss Grade 8 #18

Primary Topics: Number Sense

Secondary Topics: Factoring | Divisibility | Expressions

Answer: D

Solution:

We begin by listing, in order, the smallest 2-digit positive divisors of 630. We note that by first writing 630 as a product of its prime factors ($630 = 2 \times 3^2 \times 5 \times 7$), it may be easier to determine these divisors. The smallest five 2-digit positive divisors of 630 are 10, 14, 15, 18, and 21. The next largest 2-digit positive divisor of 630 is 30, and we notice that $21 \times 30 = 630$. That is, 21 and 30 are a pair of 2-digit positive integers whose product is 630, and they are consecutive in the ordered list of positive divisors. Thus, each of 10, 14, 15, and 18 must be paired with a divisor of 630 that is greater than 30. We may check that the divisors that pair with each of 10, 14, 15, and 18 is a 2-digit positive integer by dividing 630 by the smaller divisor.

That is, $\frac{630}{18} = 35$, $\frac{630}{15} = 42$, $\frac{630}{14} = 45$, and $\frac{630}{10} = 63$.

Thus, the pairs of 2-digit positive integers whose product is 630 are 21 and 30, 18 and 35, 15 and 42, 14 and 45, and 10 and 63, and so there are 5 such pairs.

19. At 9 a.m., Ryan had finished cutting $\frac{1}{2}$ of his lawn. At 10 a.m., he had finished cutting $\frac{7}{8}$ of his lawn. If Ryan cut his lawn at a constant rate, at what time did he finish?

- (A) 10:15 a.m. (B) 11:20 a.m. (C) 10:20 a.m. (D) 10:30 a.m. (E) 11:40 a.m.

Source: 2023 Gauss Grade 8 #19

Primary Topics: Number Sense | Algebra and Equations

Secondary Topics: Fractions/Ratios | Rates | Expressions

Answer: C

Solution:

Between 9 a.m. and 10 a.m., Ryan cut $\frac{7}{8} - \frac{1}{2} = \frac{7}{8} - \frac{4}{8} = \frac{3}{8}$ of his lawn. Ryan cut $\frac{3}{8}$ of his lawn in 1 hour (60 minutes), and so he cut $\frac{1}{8}$ of his lawn in $\frac{60}{3}$ minutes or 20 minutes. At 10 a.m., Ryan had cut $\frac{7}{8}$ of his lawn, and thus had $1 - \frac{7}{8} = \frac{1}{8}$ of his lawn left to cut. Since Ryan cuts $\frac{1}{8}$ of his lawn in 20 minutes, then he finished at 10:20 a.m.

20. A 4×4 grid is to be covered with 16 square tiles. There are four tiles in each of the colours red, black, green, and yellow. Each row must contain one tile of each colour. Each pair of tiles that touch along a side or at a corner must have different colours. In how many different ways can these tiles be arranged?
- (A) 256 (B) 24 (C) 120 (D) 576 (E) 64

Source: 2023 Gauss Grade 8 #20

Primary Topics: Counting and Probability

Secondary Topics: Counting | Probability | Graphs

Answer: B

Solution:

Begin by placing four tiles in the squares of the first row. The only restriction is that the row must contain one tile of each colour. Thus in the first row, there are 4 choices of tile colour for the first column, 3 choices for the second, 2 for the third, and 1 choice for the fourth. That is, there are $4 \times 3 \times 2 \times 1 = 24$ different ways to cover the squares in the first row using one tile of each of the four colours. For example, using R for red, B for black, G for green, and Y for yellow, the first row could contain tiles coloured $G Y R B$, in that order. Suppose that the first row does contain tiles coloured $G Y R B$, in that order. We will show that there is only one way to arrange the remaining 12 tiles in the grid.

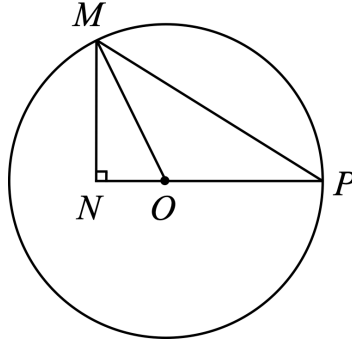
Consider the first square in the second row, that is, the square directly below the tile coloured G . The tile placed in this square cannot be coloured G since it shares an edge with the tile coloured G in row 1. Also, the tile placed in this square cannot be coloured Y since it touches the corner of the square containing the tile coloured Y in row 1. Assume that the tile in the first square of row 2 is coloured B , as shown.

Next, consider the colour of the tiles that could be placed in the second square of row 2. The tile in this square cannot be coloured Y since it shares an edge with the tile coloured Y in row 1. Also, the tile in this square cannot be coloured G since it touches the corner of the square containing the tile coloured G in row 1. Further, the tile in this square cannot be coloured R since it touches the corner of the square containing the tile coloured R in row 1. Since the first square in this row contains a tile coloured B , then we have no possible tile that can be placed in the second square of row 2. This means that the tile in the first square of row 2 cannot be coloured B , and thus it must be coloured R , as shown.

The tile in the second square of row 2 cannot be coloured Y or G (as noted earlier), and thus must be coloured B . Continuing to move right along row 2, the next tile cannot be coloured Y since it touches the corner of the square containing the tile coloured Y in row 1, and so the tile in this square must be coloured G , with the final tile in the row being coloured Y . That is, the positions of the 4 tiles in row 2 are completely determined by the tiles in row 1. Thus for each of the 24 different ways to place the tiles in row 1, there is exactly one way to place the tiles in row 2. Repeating the argument, the same is then true for the tiles in row 3 and row 4; that is, there is exactly one choice for the location of each of the coloured tiles within each of these two rows as well. The 4×4 in our example above is completed here, as shown. You should justify for yourself that each of rows 3 and 4 must contain tiles exactly as shown.

For each of the 24 different ways to cover the squares in the first row using one tile of each of the four colours, there is exactly one way to cover all remaining squares in the grid. Thus, there are 24 different ways that the tiles can be arranged.

21. In the diagram, O is the centre of a circle with radius 87, and P and M lie on the circle. N is positioned inside the circle so that PN passes through O and is perpendicular to MN .



If $MN = 63$, what is the area of $\triangle PMN$?

- (A) 3370.5 (B) 3496.5 (C) 4725.0 (D) 4630.5 (E) 4126.5

Source: 2023 Gauss Grade 8 #21

Primary Topics: Geometry and Measurement

Secondary Topics: Circles | Measurement | Area | Pythagorean Theorem

Answer: D

Solution:

Since OM is a radius of the circle, then $OM = 87$. $\triangle MNO$ is a right-angled triangle, and so by the Pythagorean Theorem, we get $OM^2 = MN^2 + NO^2$ or $87^2 = 63^2 + NO^2$, and so

$NO^2 = 87^2 - 63^2 = 3600$. Since $NO > 0$, then $NO = \sqrt{3600} = 60$.

Since OP is also a radius, then $OP = 87$, and so $NP = NO + OP = 60 + 87 = 147$. The area of $\triangle PMN$ is equal to $\frac{1}{2} \times NP \times MN = \frac{1}{2} \times 147 \times 63 = 4630.5$.

22. It took Nasrin two hours and thirty minutes to canoe the 4.5 km into her camp. Paddling much faster, the return trip took her $\frac{1}{3}$ of the time. What was Nasrin's mean (average) speed as she paddled to camp and back?
 (A) 1.25 km/h (B) 3.96 km/h (C) 1.8 km/h (D) 1.95 km/h (E) 2.7 km/h

Source: 2023 Gauss Grade 8 #22

Primary Topics: Algebra and Equations

Secondary Topics: Rates | Expressions | Equations Solving | Averages

Answer: E

Solution:

Nasrin's mean (average) speed is determined by dividing the total distance travelled, which is 9 km, by the total time. It took Nasrin 2 hours and thirty minutes, or 150 minutes, to canoe into her camp.

On the return trip, it took her $\frac{1}{3} \times 150$ minutes or 50 minutes.

Thus, the total time for Nasrin to paddle to camp and back was 200 minutes. Converting to hours, 200 minutes is 3 hours and 20 minutes, and since 20 minutes is $\frac{20}{60} = \frac{1}{3}$ hours, it took Nasrin $3\frac{1}{3}$

hours in total. Thus, Nasrin's mean speed as she paddled to camp and back was $\frac{9 \text{ km}}{3\frac{1}{3} \text{ h}}$ or $\frac{9 \text{ km}}{\frac{10}{3} \text{ h}}$,

which is equal to $9 \times \frac{3}{10} \text{ km/h} = \frac{27}{10} \text{ km/h} = 2.7 \text{ km/h}$.

23. Each of two cylinders sits on one of their circular faces on a flat surface. Cylinder A, with radius 6 cm and height 50 cm, is empty. Cylinder B, with radius 8 cm and height 50 cm, is full of water. After pouring some water from Cylinder B into Cylinder A, the height of the water in both cylinders is the same. What is the height of the water? (The volume of a cylinder with radius r and height h is $\pi r^2 h$.)
 (A) 28.1 cm (B) 25.0 cm (C) 32.0 cm (D) 44.4 cm (E) 28.6 cm

Source: 2023 Gauss Grade 8 #23

Primary Topics: Geometry and Measurement

Secondary Topics: Prisms | Measurement | Equations Solving | Decimals

Answer: C

Solution:

To begin, the volume of water in Cylinder B is $\pi \times (8 \text{ cm})^2 \times 50 \text{ cm} = 3200\pi \text{ cm}^3$. After some water is poured from Cylinder B into Cylinder A, the total volume of water in the two cylinders will be $3200\pi \text{ cm}^3$ (since no water is lost). Let h cm be the height of the water in each of the two cylinders when the height of the water in both cylinders is the same. At this time, the volume of water in Cylinder B is $\pi \times (8 \text{ cm})^2 \times h \text{ cm} = 64\pi h \text{ cm}^3$. At this time, the volume of water in Cylinder A is $\pi \times (6 \text{ cm})^2 \times h \text{ cm} = 36\pi h \text{ cm}^3$. Thus, the total volume of water in the two cylinders is

$64\pi h \text{ cm}^3 + 36\pi h \text{ cm}^3 = 100\pi h \text{ cm}^3$, and so $100\pi h = 3200\pi$ or $h = \frac{3200\pi}{100\pi} = 32$.

When the height of the water in both cylinders is the same, that height is 32 cm.

24. The number of pairs of integers a and b with $a < b$ and $a + b < 100$ satisfying the equation $\frac{a}{4} + \frac{b}{10} = 7$ is
- (A) 10 (B) 19 (C) 9 (D) 20 (E) 15

Source: 2023 Gauss Grade 8 #24

Primary Topics: Algebra and Equations

Secondary Topics: Inequalities | Equations Solving | Counting

Answer: B

Solution:

Solution 1

We begin by multiplying the given equation through by 20 to get $20 \times \frac{a}{4} + 20 \times \frac{b}{10} = 20 \times 7$, or $5a + 2b = 140$. Since $5a = 140 - 2b$ and both 140 and $2b$ are even, then $5a$ is even, which means that a is even. We start by trying $a = 20$ and $b = 20$ which is a solution, since $5 \times 20 + 2 \times 20 = 140$. This pair for a and b satisfies all of the conditions except $a < b$. We may find other solutions to the equation $5a + 2b = 140$ by adding two 5s and subtracting five 2s (this is the same as adding 10 and subtracting 10), or by subtracting two 5s and adding five 2s. Adding two 5s is equivalent to increasing the value of a by 2. Subtracting five 2s is equivalent to decreasing the value of b by 5. Consider $a = 20 + 2 = 22$ and $b = 20 - 5 = 15$. This is a solution since $5 \times 22 + 2 \times 15 = 140$. However, in this case $a > b$ and every time we add two 5s and subtract five 2s, a becomes greater and b becomes smaller. Thus, we need to go in the other direction. Consider $a = 20 - 2 = 18$ and $b = 20 + 5 = 25$. This is a solution since $5 \times 18 + 2 \times 25 = 140$. Here, $a < b$ and $a + b = 43$, so this pair satisfies all of the conditions. Next, consider $a = 18 - 2 = 16$ and $b = 25 + 5 = 30$. This is a solution since $5 \times 16 + 2 \times 30 = 140$. Here, $a < b$ and $a + b = 46$, so this pair satisfies all of the conditions. Notice that the sum $a + b$ increases by 3 on each of these steps. This means that doing this 17 more times gets us to $a = 16 - 17 \times 2 = -18$ and $b = 30 + 17 \times 5 = 115$. This is a solution since $5 \times (-18) + 2 \times 115 = 140$. Notice that it is still the case that $a < b$ and $a + b < 100$. Repeating this process one more time, we get $a = -20$ and $b = 120$, which gives $a + b = 100$ and so there are no more pairs that work. Since we know that a has to be even and we are considering all possible even values for a , there can be no other pairs that work. In total, there are $1 + 1 + 17 = 19$ pairs of integers a and b that satisfy each of the given conditions and the given equation.

Solution 2

We begin by rearranging the given equation to isolate b . Doing so, we get

$$\begin{aligned}\frac{a}{4} + \frac{b}{10} &= 7 \\ \frac{b}{10} &= 7 - \frac{a}{4} \\ 10 \times \frac{b}{10} &= 10 \times 7 - 10 \times \frac{a}{4} \quad (\text{multiplying each term by } 10) \\ b &= 70 - \frac{10a}{4} \\ b &= 70 - \frac{5a}{2}\end{aligned}$$

Since b is an integer, then $70 - \frac{5a}{2}$ is an integer, which means that $\frac{5a}{2}$ must be an integer.

Since 2 does not divide 5, then 2 must divide a and so a is even.

Since $a < b$ and $b = 70 - \frac{5a}{2}$, then

$$\begin{aligned}a &< 70 - \frac{5a}{2} \\ 2 \times a &< 2 \times 70 - 2 \times \frac{5a}{2} \quad (\text{multiplying each term by } 2) \\ 2a &< 140 - 5a \\ 7a &< 140 \\ a &< 20\end{aligned}$$

Further, since $a + b < 100$ and $b = 70 - \frac{5a}{2}$, then

$$\begin{aligned}
 a + 70 - \frac{5a}{2} &< 100 \\
 2 \cdot a + 2 \cdot 70 - 2 \cdot \frac{5a}{2} &< 2 \cdot 100 && \text{(multiplying each term by 2)} \\
 2a + 140 - 5a &< 200 \\
 -60 &< 3a \\
 -20 &< a
 \end{aligned}$$

Thus a is an even integer that is greater than -20 and less than 20 . Since there are 19 even integers from -18 to 18 inclusive, we suspect that there are 19 pairs of integers a and b that satisfy the given equation. It is a good idea (and good practice) to at least check that the largest and smallest of these values of a do indeed satisfy each of the given conditions.

When $a = -18$, we get $b = 70 - \frac{5(-18)}{2}$ or $b = 70 - 5(-9)$, and so $b = 115$. This pair satisfies the given conditions that $a < b$ and $a + b < 100$. Substituting $a = -18$ and $b = 115$ into the given equation, we get

$$\frac{a}{4} + \frac{b}{10} = \frac{-18}{4} + \frac{115}{10} = \frac{-9}{2} + \frac{23}{2} = \frac{14}{2} = 7,$$

and thus $a = -18$ and $b = 115$ is a solution.

When $a = 18$, we get $b = 70 - \frac{5(18)}{2}$ or $b = 70 - 5(9)$, and so $b = 25$. This satisfies the given conditions that $a < b$ and $a + b < 100$. Substituting $a = 18$ and $b = 25$ into the given equation, we get

$$\frac{a}{4} + \frac{b}{10} = \frac{18}{4} + \frac{25}{10} = \frac{9}{2} + \frac{5}{2} = \frac{14}{2} = 7,$$

and thus $a = 18$ and $b = 25$ is also a solution.

At this point we can be confident that for each of the 19 even integer values of a from -18 to 18 inclusive, there is an integer b for which the pair of integers a and b satisfy each of the given conditions and the given equation.

25. Given the list 2, 3, 4, 5, there are exactly three different ways to choose three integers from the list and form a triangle whose side lengths are equal to those integers. The integers chosen could be 2, 3, 4 or 2, 4, 5 or 3, 4, 5. The integers 2, 3, 5 cannot be used as side lengths of a triangle. Given the list 4, 10, 3, n , 13, there are exactly four different ways to choose three integers from the list and form a triangle whose side lengths are equal to those integers. If n is different from all other numbers in the list, then the sum of all possible values of n is
- (A) 46 (B) 29 (C) 69 (D) 23 (E) 17

Source: 2023 Gauss Grade 8 #25

Primary Topics: Geometry and Measurement | Counting and Probability

Secondary Topics: Counting | Triangles | Patterning/Sequences/Series | Trigonometry

Answer: A

Solution:

For any triangle, the sum of the lengths of two sides is always greater than the length of the third side. This property is known as the triangle inequality. If for example the side lengths of a triangle are a , b and c , then the triangle inequality says that

$$a + b > c \text{ and } a + c > b \text{ and } b + c > a.$$

We begin by considering the number of different ways to choose three integers from 3, 4, 10, 13 (without using n), and then forming a triangle whose side lengths are equal to those integers.

Consider choosing the integers 3, 4, 10. Since $3 + 4 < 10$, then it is not possible to form a triangle whose side lengths are 3, 4, 10. Consider choosing the integers 3, 4, 13. Since $3 + 4 < 13$, then it is not possible to form a triangle whose side lengths are 3, 4, 13. Consider choosing the integers 3, 10, 13. Since $3 + 10 = 13$, then it is not possible to form a triangle whose side lengths are 3, 10, 13. However, the remaining possible choice of three side lengths, 4, 10, 13, does satisfy the triangle inequality since $4 + 10 > 13$ and $4 + 13 > 10$ and $10 + 13 > 4$. Thus, without using the value of n , there is exactly one way to choose three integers and form a triangle whose side lengths are equal to those integers. This means that we need to determine values of n for which there are exactly three different ways to choose two of the integers 3, 4, 10, 13 and form a triangle whose side lengths are equal to those two integers and n . There are six possible ways to choose two integers from the list 3, 4, 10, 13. Thus for each value of n , the triangles we need to consider have side lengths: 3, 4, n or 3, 10, n or 3, 13, n or 4, 10, n or 4, 13, n or 10, 13, n . For each value of n , we need exactly three of these six triangles to satisfy the triangle inequality. Next, we use the triangle inequality to determine the restrictions on n for each of the six possible groups of triangles.

In a triangle with side lengths 3, 4, n , we get $3 + n > 4$ or $n > 1$, and $3 + 4 > n$ or $n < 7$, and $4 + n > 3$ or $n > -1$. To satisfy all three inequalities, n must be greater than 1 and less than 7. Thus, the possible values of n for which a triangle has side lengths 3, 4, n are $n = 2, 3, 4, 5, 6$. Since n must be different from all other numbers in the list, then $n = 2, 5, 6$.

In a triangle with side lengths 3, 10, n , we get $3 + n > 10$ or $n > 7$, and $3 + 10 > n$ or $n < 13$, and $10 + n > 3$ or $n > -7$. To satisfy all three inequalities, n must be greater than 7 and less than 13. Thus, the possible values of n for which a triangle has side lengths 3, 10, n are $n = 8, 9, 11, 12$ ($n \neq 10$ since 10 is in the list).

In a triangle with side lengths 3, 13, n , we get $3 + n > 13$ or $n > 10$, and $3 + 13 > n$ or $n < 16$, and $13 + n > 3$ or $n > -10$. To satisfy all three inequalities, n must be greater than 10 and less than 16. Thus, the possible values of n for which a triangle has side lengths 3, 13, n are $n = 11, 12, 14, 15$ ($n \neq 13$ since 13 is in the list).

In a triangle with side lengths 4, 10, n , we get $4 + n > 10$ or $n > 6$, and $4 + 10 > n$ or $n < 14$, and $10 + n > 4$ or $n > -6$. To satisfy all three inequalities, n must be greater than 6 and less than 14. Thus, the possible values of n for which a triangle has side lengths 4, 10, n are $n = 7, 8, 9, 11, 12$.

In a triangle with side lengths 4, 13, n , we get $4 + n > 13$ or $n > 9$, and $4 + 13 > n$ or $n < 17$, and $13 + n > 4$ or $n > -9$. To satisfy all three inequalities, n must be greater than 9 and less than 17. Thus, the possible values of n for which a triangle has side lengths 4, 13, n are $n = 11, 12, 14, 15, 16$.

In a triangle with side lengths 10, 13, n , we get $10 + n > 13$ or $n > 3$, and $10 + 13 > n$ or $n < 23$, and $13 + n > 10$ or $n > -3$. To satisfy all three inequalities, n must be greater than 3 and less than 23. Thus, the possible values of n for which a triangle has side lengths 10, 13, n are $n = 5, 6, 7, 8, 9, 11, 12, 14, 15, 16, 17, 18, 19, 20, 21, 22$.

Recall that for each value of n , we need exactly three of the six triangles to satisfy the triangle inequality (the triangle with side lengths 4, 10, 13 is the fourth). Clearly for values of n less than 7, there are too few triangles, and similarly for values of n greater than 16, there are also too few triangles. (There are at most two triangles in each of these two cases.) In the table below, we summarize our work by placing a checkmark if the triangle satisfies the triangle inequality and then counting the number of such triangles.

Therefore, there are exactly four different values of n that satisfy the given conditions, and the sum of these values of n is $8 + 9 + 14 + 15 = 46$.

